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Dynamic Load Angle Estimation Through An Angular Domain GDFT Filter For Optimal Energetic BLDC Control

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Abstract—These days, BLDC motors are increasingly used to drive speed-controlled applications (e.g., drones, air compressors, etc.) thanks to their high power density, good torque inertia ratio, and their simple square wave commutation principle. However, with growing attention towards energy usage, a more efficient sinusoidal current wave is favored over the classical square wave profile. Nevertheless, changing the waveform to absent the silence phase implicates that simple control principles (e.g., hall-sensors or back-EMF tracing) are no longer functional, and a more advanced sensorless solution is required. Earlier, an innovative sensorless load-angle-based controller was suggested with the potential to overcome all restrictions concerning state-of-the-art sensorless control. One of the proposed estimator's fundamental building blocks is a sliding discrete Fourier filter (SDFT), requiring both a signal with a fixed base frequency and a full signal period of samples before updating completely. Both requirements severely limit the application field. This paper presents an enhanced load angle estimator that can estimate the load angle 66% faster on average and without quality deterioration, regardless of the rotor speed. Both speed and accuracy of the estimated load angle are validated based on BLDC-measurements, confirming the potential and revealing possible further estimation lag decreases if a limited estimation quality regression is allowed.

Index Terms—BLDC, sensorless control, load angle estimation, GDFT.

I. INTRODUCTION

BRUSHLESS DC motors (BLDC) are gaining market share and make their entrance in more demanding and innovative applications [1]. For example, the number of air conditioners, typically BLDC powered, is expected to rise from 1.5 million today to 5.6 million by 2050, tripling the required electrical power [2]. This market growth is not surprising thanks to its high power density, good torque inertia ratio, and high holding torque. Nevertheless, to date, torque ripple reduction and energy-efficient control of BLDC motors, especially given their increasing share in global energy

consumption, remain an important issue. Traditionally, BLDC motors drive alternately inject block-wave current profiles into the stator windings, which results in a simple control principle. Already in 1994, [3] discovered that the motor's energy consumption could be reduced by improving the shape of the current profile. Hence, it better fits the back-EMF profile. Thereby, the optimal shape alters between either a basic sinusoidal and a sinusoidal with additional harmonic components. The eventual energy efficiency gain is both load and speed-dependent and can be as high as 10% [4].

Whereas a block-waved current profile only requires a discrete update of the desired current value at the so-called commutation moments, a sinusoidal profile or more complex shape requires more continuous feedback to enable optimal motor control. By the absence of the so-called silence face, feedback from, e.g., hall sensors, or a zero-crossing of the back-EMF, is no longer sufficient. Moreover, an encoder adds unnecessary cost and complexity to the system. A so-called "sensorless control algorithm," generating continuous feedback based on simple voltage and current measurements, offers a cost-effective solution. For this reason, extensive research is carried out in this field. Since the state-of-the-art in sensorless control lies with PMSMs, this will be used as a further reference in the rest of this work. The majority of the developed principles can be divided into signal injection-based [5], [6], [7] and model-based methods [8], [9], [10], [11], [5], [12]. The first group uses a high-frequency injected signal to estimate the rotor position, while the second group uses a self-correcting control loop on top of a mathematical motor model. Although both principles have proven to work, practical limitations still hamper the implementation in the cost-critical BLDC market. Signal injection works well up to mid-speed ranges but is limited at higher speeds by the maximum inverter output frequency [5]. Moreover, the injected harmonics in the stator windings cause additional energy losses, torque ripple and unwanted acoustic noise [6]. Finally, signal injection places additional demands on the motor construction [13], which goes against the cost-critical market in which BLDC's are used. On the other hand, the observer-based methods do not require additional investment in hardware and function well starting from mid-speed range but become unstable and thus less reliable at lower speeds [14]. Moreover, to control the motor dynamically enough, difficult-to-estimate application-dependent parameters such as damping and inertia are required, which complicates the applicability [15], [8]. As summarized by [8], both control principles remain

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vulnerable to external disturbances despite continuous efforts to improve control performance.

To date, the literature lacks the description of an energy-efficient, robust BLDC torque controller that is as easily implementable as the classic BLDC algorithm. A new innovative control principle was recently proposed based on the estimation of the load angle, being the angular displacement between the stator current vector and the rotor flux vector [16]. This work introduces the following essential enhancements to [16]:

- 1) One of the fundamental building blocks of the proposed load angle estimator is the filter used to obtain the amplitude and phase of the fundamental wave of voltage and current. So far, the discrete Fourier filter [17] used for this purpose typically provides accurate harmonic information about the fundamental wave yet requires a fixed time base or sample window. However, a changing rotor speed implies a change in the ground wave's signal period, leading to a temporary disturbance in estimating the amplitude and phase during a velocity transient and the next signal period, resulting in unreliable load angle information. Estimating the load angle at variable speed is impossible up to now, which severely limits the use of the load angle controller. Therefore, based on the renewed control structure, this work proposes a sampling strategy in the angular domain, allowing the dynamic filtering of voltage and current at continuously variable speed in the frequency domain. This enables continuous estimation of the motor's load angle and an improved optimal control.
- 2) All samples after a sudden transient in speed, load, or current will contribute to the update of the sliding discrete fourier transform (SDFT) to take the transient into account. However, pre-transient behavior will only stop influencing the SDFT result after one full signal period of samples. To substantially improve the dynamics of the load angle estimator, the recently developed generalized discrete Fourier filter (GDFT-filter) [18] is introduced in this work. Yet, the filter still needs tuning, taking the trade-off between estimation accuracy and dynamicity into account. Therefore, this work proposes a BLDC-tuned GDFT-filter, reducing the estimation lag by 66,66% on average and thus severely improving the load angle estimation accuracy.

The remainder of this work is structured as follows: chapter II discusses BLDC torque generation and control, and its relation to the load angle δ . Next, chapter III introduces the GDFT filter and its configuration, starting from the well-known SDFT filter. Chapter IV deals with the estimation of the load angle in the angular domain. Since the GDFT configuration depends on the frequency content of voltage and current, measured BLDC data is analyzed in chapter V to validate the theoretically derived harmonic spectrum from III. Finally, chapter VI validates the performance of the newly proposed estimator based on transient BLDC measurement data.

II. TORQUE GENERATION AND CONTROL

A. Torque generation

For the concept of load angle estimation, it is essential to consider the generated electromagnetic torque

$$T_{generated} = |i_s| \cdot \sin(\delta_s) \cdot \psi_r \quad (1)$$

with $|i_s|$ beeing the resultant stator current vector magnitude, δ_s the load angle and ψ_r the magnetic rotor flux. Following (1), torque generation is most energy-optimal when the stator current i_s is controlled so that the load angle reaches 90° . Though the angle cannot be measured directly online. Thanks to Lenz's law the resultant angle current vector $\angle i_{ab}$ and back-EMF vector $\angle e_{ab}$ can be related to the load angle [19] ensuing:

$$\delta_s = \pi - (\angle e_{ab} - \angle i_{ab}) \quad (2)$$

The stator current angle $\angle i_{ab}$ can easily be determined by measuring current in the stator windings. The back-EMF angle $\angle e_{ab}$, on the contrary, cannot be measured directly, but can be described as a function of the line voltage v_{ab} over and current i_{ab} through the winding with resistance R_a and induction L_a :

$$e_{ab}(t) = v_{ab}(t) - R_a i_{ab}(t) - L_a \frac{di_{ab}(t)}{dt} \quad (3)$$

However, due to the presence of noise, the derivative in (3) is distorted, so it is wiser to describe the back-EMF in the frequency domain:

$$E_{ab}(j\omega) = U_{ab}(j\omega) - R_a I_{ab}(j\omega) - j\omega L_a I_{ab}(j\omega) \quad (4)$$

Since only the ground wave with pulsation ω_1 contributes to the average torque, all higher frequency harmonics are ignored [20]. Moreover, (4) shows that the load angle estimation solely depends on the stator resistance, inductance, and the dynamic filtering of voltage and current. Estimating R and L has matured considerably over the past decades due to its multitude of described methodologies [21]. The estimation of a signal's amplitude and phase is described in the literature by methods that can be divided into frequency and time domain-dependent solutions [22]. Analysis in the frequency domain is mainly associated with the Fourier transformation, which mathematically isolates the desired harmonic. They are known for their simplicity, selectivity, and high accuracy but exhibit a relatively slow dynamic response and require synchronization between the sample and signal frequency. A time-domain analysis is usually more dynamic and less computationally demanding than a frequency-domain analysis, but often less accurate, regularly requires case-specific tuning, and usually estimates the amplitude and phase separately. [23]

B. Torque control

In literature, general sensorless BLDC control structures typically estimate the rotor position first, after which vector control regulates the generated torque and speed. However, the proposed load angle estimator does not provide information about the rotor angle's exact location, excluding classical control structures. To overcome this hurdle, [16] earlier suggested a new structure shown in figure (1), regulating the motor's performance through the estimated load angle $\hat{\delta}$. This is done

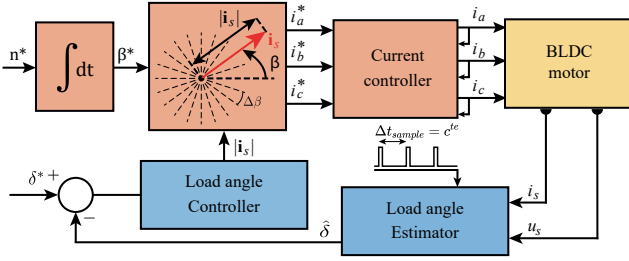


Fig. 1: Previously developed [16] control structure to regulate a BLDC motor through a load angle estimator and controller. Sampling of voltage and current happens with a constant time interval.

by reversing the control principle and imposing the rotor speed in open-loop instead of estimating it. In essence, the load angle δ is used to close the control loop. A standard PI controller manipulates the stator current amplitude setpoint $|i_s^*|$ to control the load angle δ .

From equation (1), it was previously deduced that the motor is the most energy-efficient when the load angle δ equals $\pi/2$ rad. However, [24] elucidated that this optimum is an unstable operating point, and to guarantee stable operation, the load angle setpoint δ^* should preferably be chosen lower. Based on [24], it can be concluded that a reduced estimator lag results in a smaller safety margin to the theoretical optimum, and eventually, a more energy-efficient drive. To reduce the estimator lag, the Fourier transform is considered in more detail.

III. ENHANCED ESTIMATION OF THE FUNDAMENTAL AMPLITUDE AND PHASE

A. Conventional estimation based on the SDFT

The discrete Fourier transformation (DFT) mathematically converts a finite sequence of evenly distributed samples from the non-frequency domain (e.g., the time or angle domain) to the discrete frequency domain. The frequency spectrum of a discretely sampled signal is described according to:

$$X^k(n) = \sum_{i=n-N+1}^n \left[x(i) e^{-j2\pi \frac{i}{N} k} \right] \quad (5)$$

with $k = 0, \pm 1, \dots, \pm \left(\frac{N}{2}-1\right)$ being the harmonic order, X^k the complex DFT result of the k -th harmonic after taking the n^{th} input sample, $x(i)$ the input signal at sample i and N the total number of sequence samples per signal period. Implementation of the DFT algorithm on a real-time embedded system is due to the required $O(N^2)$ operations computationally demanding. Therefore, often the computationally more efficient sliding DFT (SDFT) is implemented which in $O(N)$ operations gives the complex result $X^k(n)$ for the k^{th} harmonic on index n calculated according to:

$$X^k(n) = \left[X^k(n-1) - x(n-N) + x(n) \right] e^{j2\pi \frac{k}{N}} \quad (6)$$

The SDFT filter only uses the frequency domain signal component $X^k(n-1)$ determined during the previous cycle, the

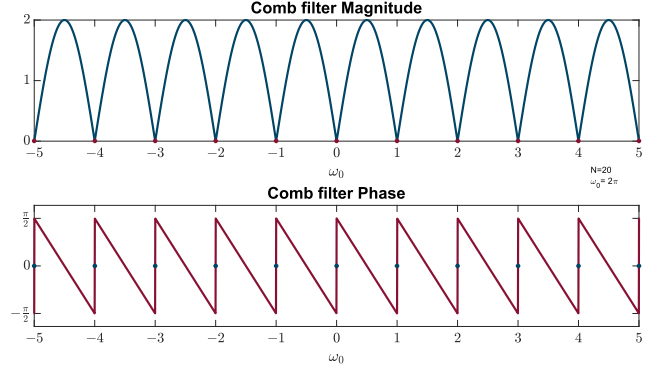


Fig. 2: The SDFT comb filter magnitude and phase response.

newly added sample $x(n)$, and the oldest sample $x(n-N)$ from the sample window.

Retrieving the instantaneous value of the k -th harmonic to index n in the original non-frequency domain, e.g. the time domain, is possible using the inverse DFT (iDFT):

$$x^k(n) = \frac{1}{N} X^k(n) e^{j2\pi \frac{kn}{N}} \quad (7)$$

However, in-depth analysis of the SDFT filter's properties in the frequency domain based on equations (6) and (7) is complicated due to the complex exponential formulation. To better understand the working principle of the SDFT, as in [18], the response from the input signal sequence $x(n)$ to the output signal sequence $x^k(n)$ for a certain harmonic k can be transformed to the Z-domain following:

$$\begin{aligned} G_{SDFT}^k(z) &= \frac{Z \left[x^k(n) \right]}{Z \left[x(n) \right]} = \frac{1}{N} \frac{1 - z^{-N}}{1 - e^{j2\pi \frac{k}{N}} z^{-1}} \\ &= \underbrace{(1 - z^{-N})}_{H_c(z)} \underbrace{\frac{1}{1 - e^{j2\pi \frac{k}{N}} z^{-1}}}_{H_r^k(z)} \frac{1}{N} \end{aligned} \quad (8)$$

The transfer function $G_{SDFT}^k(z)$ can be rewritten and split into a cascade structure consisting of three elements: the comb filter $H_c(z)$, the complex resonator $H_r^k(z)$ and the magnitude correction term λ . As shown in figure 2, the frequency response of the comb filter $H_c(z)$ is a typical FIR equiripple lobe structure with zero gain at each k^{th} harmonic, eliminating all harmonic content within the signal. The filter's phase is a saw tooth with a neutral phase shift for every k^{th} harmonic. From the further expansions of the comb filter:

$$H_c(z) = 1 - z^{-N} = \prod_{n=0}^{N-1} \left(1 - e^{j \frac{2\pi}{N} n} z^{-1} \right) \quad (9)$$

it can be deduced that the filter introduces N evenly distributed zeros on the unit circle at positions $z = e^{j \frac{2\pi}{N} n}$, as shown in blue in figure 3. These zeros cause the specific harmonic filter behavior of the comb filter under the condition that the sample window $N \cdot T_{sample}$ is exactly equal to the ground harmonic T_{signal} .

The secret behind isolating and extracting only the desired k^{th} harmonic lies in the single-pole complex resonator $H_r^k(z)$,

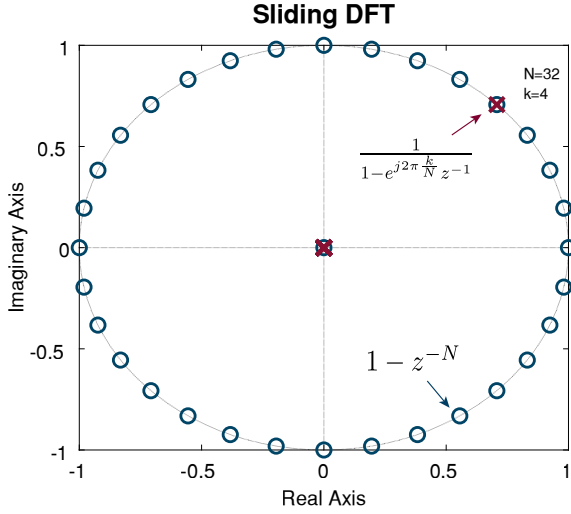


Fig. 3: Pole zero diagram of a conventional SDFT filter.

causing a zero-pole cancellation at the desired harmonic frequency. The resonator position, as shown in figure 3 in red, is a compensating pole on the unit circle at exactly one of the harmonic zeros introduced by the comb-filter. The pole neutralizes that zero's effect so that the content on the corresponding frequency is no longer filtered out but passed through after correction by the gain factor λ . The SDFT output is a complex value containing the magnitude and phase of the desired harmonic. From the frequency response of the cascade of $H_c(z) \cdot H_r^k(z) \cdot \lambda$ shown in figure 4 it can be deduced that for $k = 3$ the SDFT realizes a unit gain without phase shift for the third harmonic, while all other k^{th} -harmonics are neutralized.

In brief, the frequency-selective property of the DFT filter relies on the zero pole cancellation. For this purpose, the comb-filter introduces N evenly distributed zeros over the unit circle, filtering out all k^{th} -harmonic frequencies up to the $\frac{N}{2} - 1$ order. However, introducing N zeros on the unit circle implies a $N \cdot T_{sample}$ time delay of the output signal. As a result, an amplitude change at the filter's input is only fully addressed by the comb filter N samples later. Improving the response speed by increasing the sampling frequency is not a solution since the number of introduced zeros would proportionally increase so that the total time required to correct for a change remains the same. The only way to improve the SDFT speed is by altering the buffer's size, i.e., the number of zeros N introduced to compensate for present harmonic content. Luckily, most signals lack a frequency spectrum consisting, for example, of both even and odd harmonics spread across the entire spectrum, making the introduction of all N zeros for filtering these harmonics superfluous and thus the filter unnecessarily slow.

B. From sliding DFT to the accelerated 'generalized' DFT

As pointed out earlier, the dynamics and filtering properties of the (S)DFT are directly linked to the construction of the comb filter $H_c(z)$. By intervening in the filter structure, the recently introduced 'Generalized DFT' tries to accelerate the

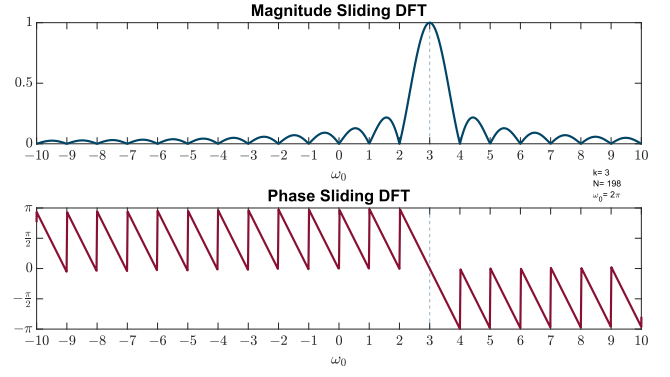


Fig. 4: Phase and magnitude response of a SDFT filter with complex resonator at $k = 3$.

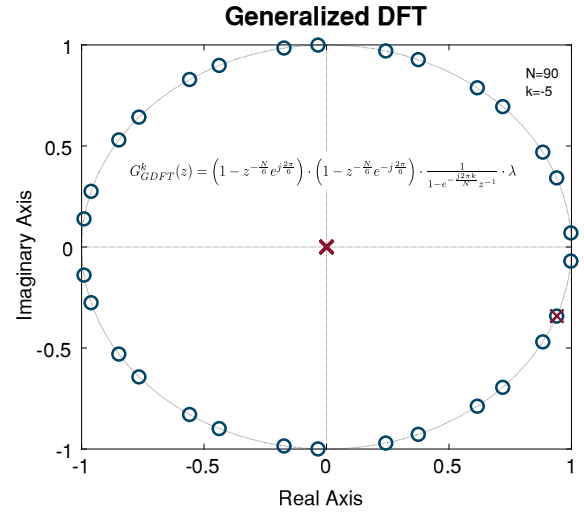


Fig. 5: Pole zero-diagram of a GDFT filter comprising a serial comb filter with $m = 6$, $l = \pm 1$, a complex resonator at $k = -5$ and a magnitude correction.

estimation of the desired k^{th} harmonic [18]. Though, the critical structure of the proposed filter remains identical to that of a classic DFT filter. The overall idea is to filter more dynamically by no longer neutralizing all k harmonics by introducing N -zeros on the unit circle. Instead GDFT only compensates for the notable harmonics by introducing targeted zeros according to a harmonics scenario following $h = m \cdot b + l$ range, where m and l are integers, $m > 0$ and $b = 0, \pm 1, \dots, (N-1)$. Now, by substituting $m \cdot b + l$ in the zero locations of equation (9), the locations of the reconstructed comb's zeros are determined:

$$H_c(z) = \prod_{b=0}^{N-1} \left(1 - e^{j\frac{2\pi}{N} (mb+l)} z^{-\frac{1}{m}} \right) = 1 - z^{-\frac{N}{m}} e^{j\frac{2\pi l}{m}} \quad (10)$$

Equation (10) indicates that the comb filter no longer introduces a delay with factor N , but $\frac{N}{m}$, thereby improving the dynamics of the filter by a factor m . Stated differently, the GDFT filter needs m times fewer samples to extract the same frequency content from the input signal as SDFT, making the filter m times faster and thus more dynamic. On the other

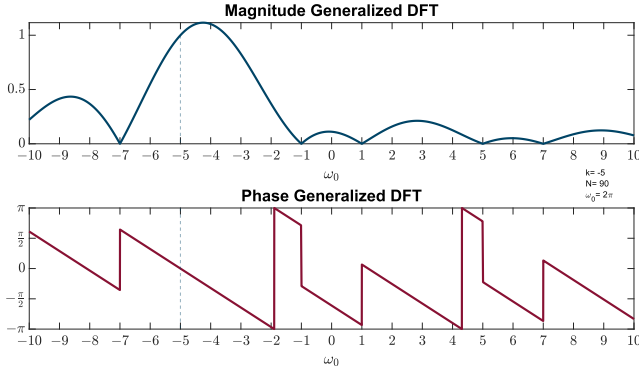


Fig. 6: Phase and magnitude response of a GDFT filter comprising a serial comb filter with $m = 6$, $l = \pm 1$, a complex resonator at $k = -5$ and a magnitude correction.

hand, the proposed comb filter's $H_c(z)$ biggest drawback is that only those harmonics corresponding to the well-defined set of $h = m \cdot b + l$ are neutralized, making it more noise sensitive. Complicated harmonic scenarios that are difficult to comprehend in a single sequence can be purged by sequencing different comb-filters following

$$H_c(z) = \left(1 - z^{-\frac{N}{m_1}} e^{j\frac{2\pi l_1}{m_1}}\right) \left(1 - z^{-\frac{N}{m_2}} e^{j\frac{2\pi l_2}{m_2}}\right) \dots \left(1 - z^{-\frac{N}{m_p}} e^{j\frac{2\pi l_p}{m_p}}\right) \quad (11)$$

The final delay induced by the concatenation of multiple filters is equal to the sum of the partial delays caused by each individual comb filter: $\sum_{r=1}^p \frac{N}{m_r}$.

In summary, the critical element in filtering, isolating, and reconstructing the desired harmonic remains, as with SDFT, the zero-pole cancellation, so only one specific harmonic frequency can pass through. The magnitude correction on the other hand does need an update to correct the attenuation and phase shift caused by the modified comb filter. In [18] it was shown that for a randomly constructed comb filter the correction term λ for the k^{th} harmonic can be obtained according to

$$\lambda = \frac{e^{-j2\pi \frac{k}{N}}}{\left[H_c(z)\right]_{z=e^{j2\pi \frac{k}{N}}}} \quad (12)$$

with $\left[H_c(z)\right]'$ being the derivative from $H_c(z)$ to Z so that:

$$G_{GDFT}^k(z) \Big|_{z=e^{j2\pi \frac{k}{N}}} = \lambda H_c(z) H_r^k(z) \Big|_{z=e^{j2\pi \frac{k}{N}}} = 1 \quad (13)$$

C. Theoretical BLDC voltage and current frequency spectrum

BLDC motors' torque generation is most optimal when the desired current profile evolves sinusoidally. However, because of, among other things, the voltage fed inverter bridge and its switching strategy (time-harmonics) [20], and the physical construction of the motor (spatial-harmonics) [25], [26], both the final voltage across and the current through the stator windings is harmonically polluted. And since the estimation delay of the proposed GDFT filter directly depends on the harmonics present and the related design of the comb filter

$H_c(z)$, prior knowledge of the expected harmonic content in both signals is indispensable.

In the literature, both time and spatial harmonic scenarios are described analytically. For example, it is expected that the harmonic content in the output voltage of a 3 phase multi-pulse PWM-controlled inverter depends on both the frequency of the carrier wave and the modulation wave. On the other hand, the complexity of the harmonic scenario depends on the motor's structural properties. For the fundamental time-harmonic order, [26] concluded that for the simplest variant, a three phase machine with an integer number of slots per pole and per phase, only odd harmonics will be present, excluding the 3rd harmonic.

The other way around, by looking at which harmonics are certainly not expected, it can easily be shown that for a three-phase machine, the theoretical frequency response of the measured voltage and current will not contain both 2^{nd} and 3^{rd} harmonics with corresponding multiples. All the even harmonics should be missing because of the symmetric nature of the desired (sinusoidal) current, whereas the 3^{rd} harmonic neutralizes itself due to the 120° phase shift of the star-connected motor windings [20]. To conclude, only the 5^{th} , 7^{th} , 11^{th} , 13^{th} , ..., harmonics are expected.

D. Theoretical optimal design of a BLDC GDFT filter

To neutralise the harmonic BLDC scenario predicted in III-C, ideally two comb filters are serialised with m -factor = 6 and $l = \pm 1$, filtering harmonics according to $h_1 = +1, +7, -5, +13, -11, \dots$ and $h_2 = -1, +5, -7, +11, -13, \dots$. Figure 5 illustrates the pole-zero diagram of this composite filter. The fact that the zeros are no longer proportionally distributed over the unit circle makes it clear that no longer all harmonics orders are filtered out. The total time delay introduced by the proposed serialized comb filters is equal to $\frac{1}{3}$ of the fundamental cycle time.

Moreover, figure 6 shows the spectral phase and magnitude properties of the proposed BLDC comb filter with λ corrected unit gain passage and zero phase shift of the -5^{th} harmonic. The limited spectral selectivity exposes that in the presence of undesirable unanticipated harmonic content, e.g., $\pm 3^{th}$ order harmonic content, errors may arise on the amplitude and phase estimate. Sometimes, noise is even amplified locally, contributing to a potentially more significant estimation error. Therefore, in the remainder of this work, attention should be paid to the eventual present harmonic signal content and related comb filter design to minimize an erroneous estimation.

IV. CONTINUOUS ESTIMATION OF THE LOAD ANGLE AT VARIABLE ROTATION SPEED

Like SDFT, GDFT also requires a whole number of equidistant samples distributed along the sample window to avoid spectral leakage. In the simplest scenario, the motor runs at a known constant rotation speed ω_{mech} and, additionally, the sample time t_{sample} is a whole divisor of the fundamental signal period T_{signal} which makes it easy to split the latter into N equal parts of time. However, this would limit the possible rotational speeds to ensure that N is an integer. Moreover, as

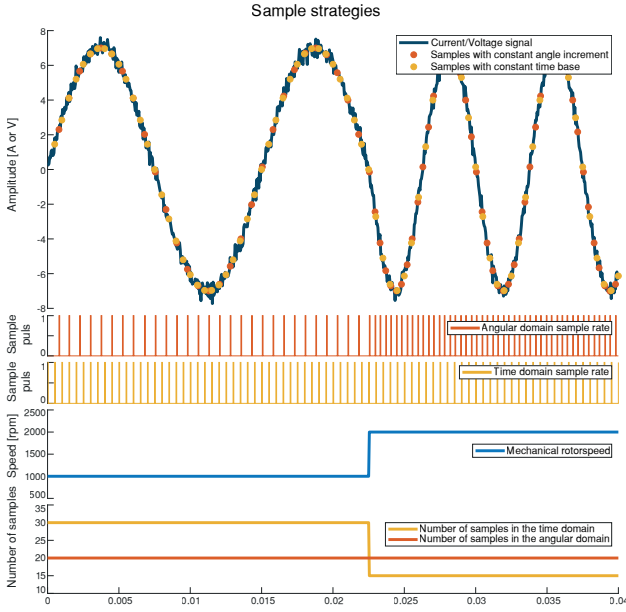


Fig. 7: Time and angular domain based sampling strategies.

shown in figure 7, sampling on a constant time basis implies that with a changing rotor speed, the number of samples per signal period also changes. GDFT cannot fundamentally deal with this, which makes that during a speed transient and the next $\frac{N}{m}$ samples, the algorithm cannot estimate the load angle reliably. To ensure continuous feedback within the existing control structure, replacing the sampling strategy in the time domain with a strategy with constant increment in the angular domain is proposed. This way, the number of samples per electrical signal period remains fixed so that the Fourier filter no longer has to 'reset' with changing rotor speed. As a result, continuous load angle information is available independent of the rotor speed. Moreover, sampling in the angular domain is advantageous for the embedded implementation because of the fixed allocated memory. As a rule of thumb, it is assumed that $N = 20$ samples are sufficient to reconstruct the 1st harmonic [27]. A sampling strategy in the time domain causes a changing number of samples so that, depending on the ratio between the minimum and maximum motor speed, soon, 30 times more allocated memory is required [28].

A. Implementation of an angle-based sample methodology within the existing load angle control structure.

The proposed control structure [16], shown in figure 1, imposes the rotation speed of the magnetic stator field in open-loop. Additionally, the load angle δ is estimated and used to control the motor. Practically, a rotating magnetic field is generated by repositioning the desired stator current vector i_s over an angle $\Delta\beta$ per time step ΔT . The size of the angular displacement is chosen independently of the cycle time ΔT , which is fixed. The higher the speed, the larger the necessary angle increment $\Delta\beta$ per time step. By subsequently selecting the time step linked to the cycle time of the controller $T_{controller}$ sufficiently small (up to $5 \mu s$ [28]), a

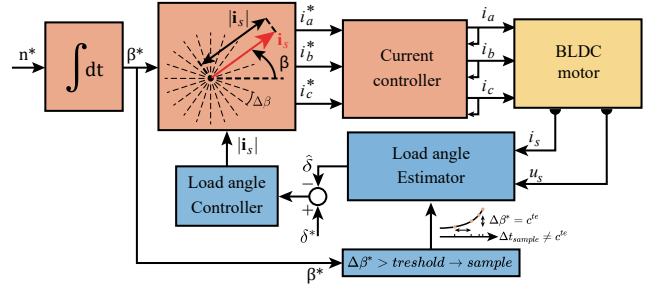


Fig. 8: Newly proposed control structure with sampling strategy in the angular domain.

smoothly rotating field is provided. In the old control structure [16], sampling is performed after each fixed number series of x clock cycles. In the new control structure, schematically represented in figure 8, imposing the rotor speed is retained while both voltage and current are sampled with constant angular increment $\Delta\beta$ across the full speed range.

V. STATIC AND DYNAMIC ANALYSIS OF THE VOLTAGE AND CURRENT FREQUENCY SPECTRUM, AND ITS IMPACT ON THE USE OF THE PROPOSED GDFT FILTER

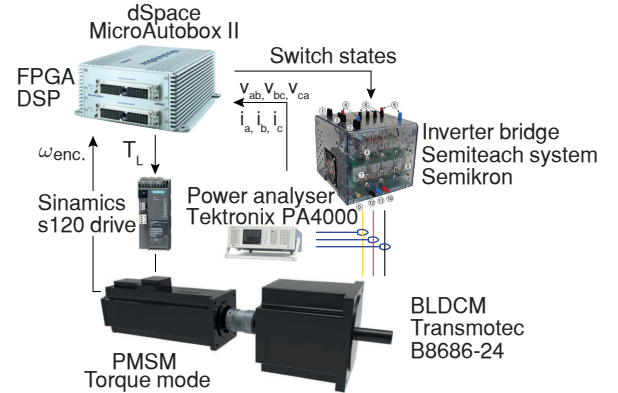


Fig. 9: Test setup used for measuring the BLDC's voltages and currents under specific static or transient test conditions.

TABLE I: Main properties of the BLDC motor under test

BLDC Transmotec B8686-24					
U_{nom}	I_{nom}	η_{nom}	T_{nom}	R_{ab}	L_{ab}
24 V	13,76 A	3062 rpm	0,703 Nm	0,091 Ω	0,23 mH

As demonstrated earlier, the estimator dynamics' achievable improvements by estimating the load angle with GDFT instead of SDFT depends on the present harmonic content in the measured voltage and current signal. Therefore, analysis is carried out under steady-state conditions at different speeds and motor loads. A transient harmonic analysis repeats this on both a speed step at constant load and a load step at a constant rotation speed. All measurements and experiments

Voltage frequency spectrum analysis

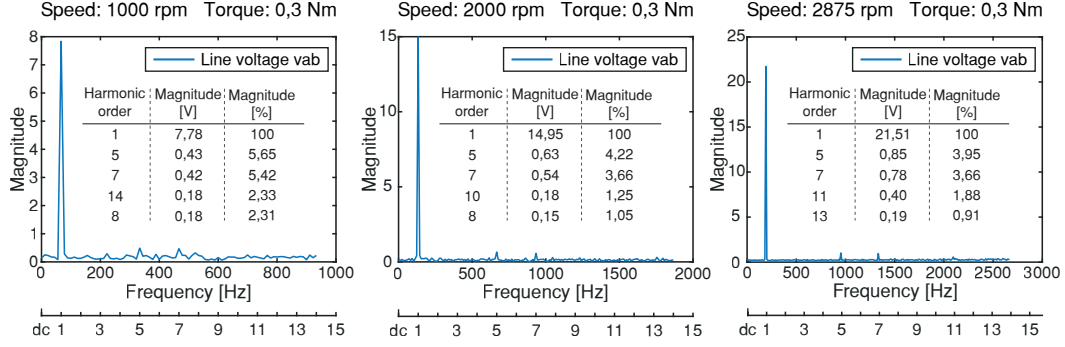


Fig. 10: Frequency analysis of line voltage at 0.3Nm and 1000 rpm (left), 2000 rpm (centre) and 2875 rpm (right). Per speed the 5 harmonics with the largest absolute value are displayed.

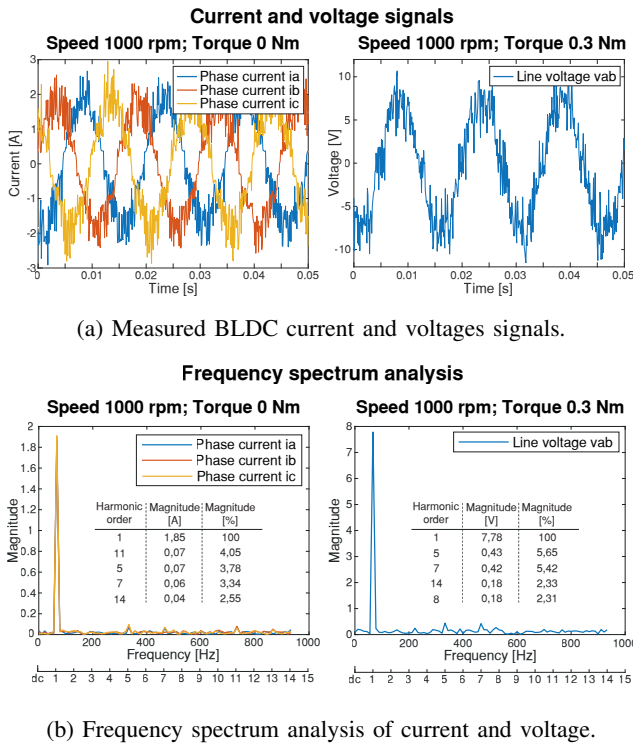


Fig. 11: Analysis of the speed-torque combinations for voltage or current showing the relatively highest harmonic values.

below are conducted using the test setup shown in figure 9, consisting of a BLDC Transmotec B8686-24 motor as the driving force and a PMSM motor with a commercial drive as the load motor. The BLDC is controlled using an inverter bridge and programmable dSpace-FPGA. The properties of the BLDC under test are summarised in table I.

A. Static harmonic analysis

Both voltage u_{ab} and current i_{ab} were analyzed at three constant rotation speeds (1000 rpm, 2000 rpm, and 3000 rpm) at no-load and at a load torque with a setpoint of 0,3Nm. Each signal was captured for 0,1s at a sampling frequency of 10 kHz. Figure 10 visualizes the v_{ab} frequency spectrum

at the three aforementioned rotation speeds, consistently at 0,3Nm. Generally speaking, the spectrum analysis confirms the findings of chapter III-C. The odd harmonics, except for the 3rd, are dominant. More specifically, the 5th and 7th order harmonics are the most impactful. Surprisingly, at lower rotational speeds, remnants of even harmonics can be traced more significantly than the expected 11th and 13th order harmonics. At no-load, a similar story is recurring. Moreover, in figure 11 both the voltage and current frequency spectrum are shown for the torque-speed combination with the percentage relatively highest harmonic magnitude. Striking is the prominent presence of the 11th harmonic in the current at no load and its absence in the voltage signal at 0,3Nm. Therefore, a high magnitude in a specific harmonic presence in, e.g., the voltage, does not directly lead to the same high magnitude in the current spectrum.

Secondly, the signal-to-noise (SNR) ratio is a meaningful indicator as noise negatively influences the estimation accuracy. Therefore, the ratio can be used to get insights into noise behavior through the speed and torque range, thus marking ideal test circumstances. Table II summarizes the SNR for both voltage v_{ab} and current i_a under all considered scenarios. Generally, the SNR is worst at the lowest rotational speed, whether at no load or not, which is also visually apparent in figure 10. The voltage v_{ab} SNR is mostly influenced by the rotation speed rather than the torque, and increases roughly from 10dB to 20dB. Opposite, the current SNR is only slightly influenced by a changing speed, yet even more by a varying torque.

Consequently, from both the analysis of the harmonic's order magnitude and the SNR, it can be deduced that lower speed estimations suffer the most from unexpected harmonic content and overall signal noise. Further analysis in section VI will therefore focus on the evaluation of the renewed load angle estimator at those low speeds to give a definitive answer on the optimal GDFT's redesigned comb filter and related gain.

B. Transient harmonic analysis

Since most BLDC applications require continuously varying speed and torque adjustments, transient analysis of the present

TABLE II: Signal to noise ratio of measured current and voltage signals

	1000 tr/min		2000 tr/min		3000 tr/min	
	Current [dB]	Voltage [dB]	Current [dB]	Voltage [dB]	Current [dB]	Voltage [dB]
No-load	12,09	13,99	11,09	18,23	12,29	19,22
0,3 Nm	19,52	9,5	19,96	14,71	21,44	17,77

harmonic content seems appropriate. The analysis is performed based on a speed or torque step change with a respective magnitude of 200 rpm or 0.2Nm, employing a spectrogram with a sampling frequency of 10kHz, resolution of 20Hz and overlap of 99%. Moreover, all transient measurements are performed in an open-loop to eliminate controller influences and maximize possible variations of currents and voltages. Because of lower non-harmonic noise, the transient analysis is performed at high rotational speed to make unforeseen harmonic effects more easily visualized.

Figure 12 shows the time-frequency space of voltage v_{ab} for a speed jump from 3000 to 3200 rpm at no-load. Due to the sudden speed increase, the required torque also suddenly increases. Given the open-loop character, this manifests itself in an increase of the load angle, causing the speed to attenuate around its new operating point. On the spectrogram, these fluctuations are reflected in the ground wave and the 5th, 7th, and 11th harmonics, with the amplitude of the frequency deviation increasing as the harmonic order increases. As a result, more signal content will pass the comb filter at higher harmonic order. On the other hand, the spectrogram nicely shows no other unexpected harmonic effects occurs within the transient.

Figure 13 shows a similar analysis for the line voltage upon a load torque jump at $t=0,3s$ from 0 Nm to 0,2 Nm. Again, both the load angle and the rotor speed will fluctuate around their new operating point. Alongside the usual harmonic content's apparent presence, the appearance of random white noise across the entire frequency spectrum is noticeable.

VI. EXPERIMENTAL VALIDATION OF THE NEW LOAD ANGLE ESTIMATOR BASED ON REAL MEASURED DATA

From section III-C, it is apparent that BLDC motors have a harmonic scenario for which a comb filter with m equal to 6 is theoretically ideal. Nevertheless, the frequency analysis of voltage and current in section V-A showed that unforeseen even harmonic content could be present in the lower speed regions. Besides, the SNR exposed that with decreasing frequency, relatively more signal noise is present. Consequently, the question arises as to whether the theoretically determined m -factor in III-D is the most optimal GDFT configuration.

A. Validation principle

To validate the improved load angle estimator described in III, multiple candidates comb-filter designs ($m = 4, 6, 8$ or 10) are subject to measured BLDC current and voltage signals obtained with the testbench shown in figure 9. By comparing the measured δ_m and the estimated load angle $\hat{\delta}$, both the GDFT filter's impact and the sampling strategy in the angular domain are evaluated. The performance of the proposed estimators is expressed in terms of the accuracy and

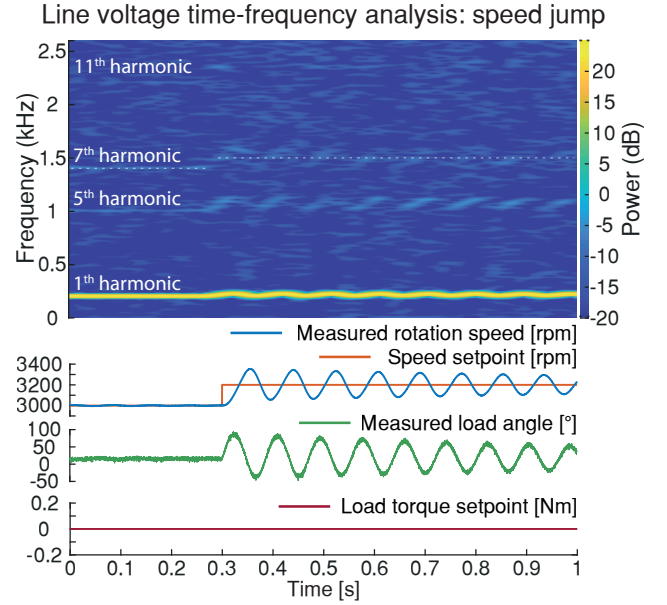


Fig. 12: Dynamic frequency analysis of the line voltage after a speed jump at $t = 0,3s$ under no-load.

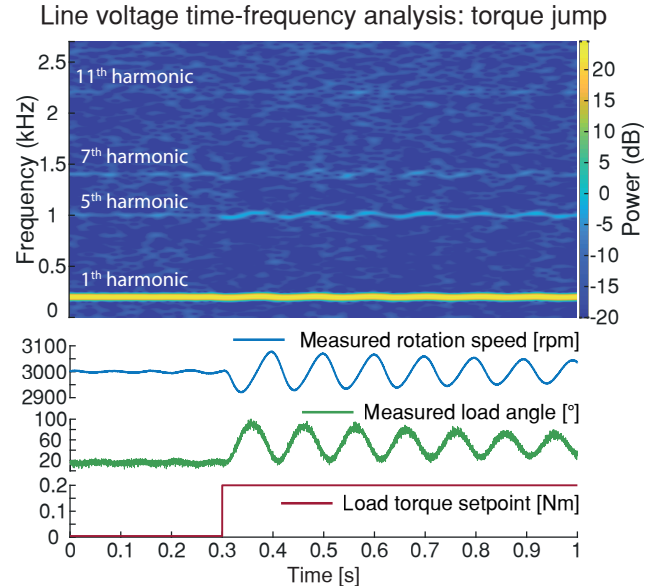


Fig. 13: Dynamic frequency analysis of the line voltage after zero load torque jump to 0,2Nm at constant speed ($\pm 3000rpm$) at $t = 0,3s$.

stability of the estimate, and the average speed gain. Validation of the performance of the novel estimator is carried out in two different experiments in open-loop. First, a torque step is used at constant rotor speed to analyze the impact of multiple GDFT comb-filter designs on the load angle estimation error. Second, upon a speed jump, the combined effect of each of the different GDFT filters with a sampling strategy in the angular domain is analyzed. For both analyses, the SDFT [16] is used as a reference. To neutralize both the influence of R and L on the estimation result, both parameters' exact values were calculated using encoder rotor angle information.

1) *Evaluation of the accuracy and stability of the estimation:* First of all, the overall estimation accuracy is an important performance indicator. Starting from the instantaneous measured load angle δ_m and the estimated load angle $\hat{\delta}$, for each comb filter configuration, an instantaneous error value ϵ is calculated following

$$\epsilon_{error} = \hat{\delta} - \delta_m \quad (14)$$

Based on (14), three key numbers, namely the mean error value μ , the standard deviation σ , and the SNR, are calculated along the transient period to interpret the estimator's transient performance. Both the mean error μ and standard deviation σ indicate the estimator's accuracy and stability over the complete considered transient window and can be seen as 'overall' performance factors. The SNR serves as a relative basis of comparison for short-term fluctuations/noise, thus indicating the stability of the estimator.

2) *Evaluation of the percental speed gain:* Besides the estimation accuracy, it is interesting to consider the eventual speed gain of the estimator itself. The theoretically derived speed gain of 66,67% from III-C is validated by calculating the timeshift between the measured and each of the estimated load angle signals using a cross-correlation $\hat{R}$. The lag between the measured and one of the estimated load angles is the shift with the highest correlation value. Calculating the correlation between two sequences a_n and b_n , with $-\infty < n < +\infty$ is possible according to [29]

$$CrossCorrSeq(i) = \hat{R}_{ab}(i - N) \quad (15)$$

with $i = 1, 2, \dots, 2N - 1$, and

$$\hat{R}_{ab}(i) = \begin{cases} \sum_{n=0}^{N-i-1} a_{n+i} b_n^* & i \geq 0 \\ \hat{R}_{ba}^*(-i) & i < 0 \end{cases} \quad (16)$$

where $*$ is the complex conjugate. Moreover, based on these time shifts, the percentage speed gain of each considered GDFT estimator with varying m-factor is determined with respect to the classical SDFT estimator following

$$Time\ shift\ gain[\%] = 100 - \frac{Time\ shift_{GDFT;m}}{Time\ shift_{SDFT;ref}} \cdot 100 \quad (17)$$

B. Validation of the influence of the comb filter m-factor on the load angle estimation

For validation, a torque jump of 0,2Nm was applied to the load motor on the test bench, while voltage, current, and the

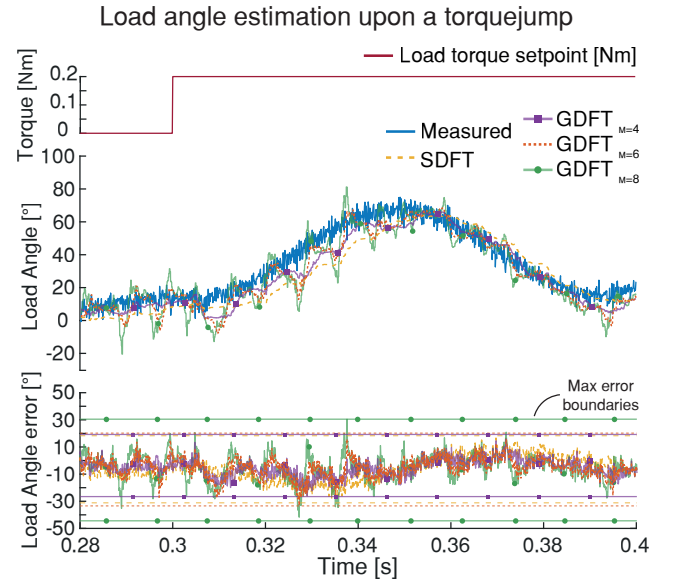


Fig. 14: Analysis of the estimated load angle $\hat{\delta}$ after a torque jump on the basis of a SDFT-filter and the GDFT structure with variable m-factor. The lower section shows the instantaneous estimation error ϵ between the measured and estimated angle for each estimator. Additionally, horizontal maximum extremity lines are added, indicating the maximum load angle error.

rotor angle were continuously measured for 0,7s. Next, both signals were filtered through a GDFT filter with a different configuration ($m = 4, 6, 8$ or 10), after which, based on equations (2) and (4), the load angle was estimated. Figure 14 zooms in on the first 100 ms after the step response from figure 13, and displays both the real measured δ_m and estimated $\hat{\delta}$ load angles.

1) *Evaluation of the accuracy and stability of the estimation:* From figure 14, it is visually clear that, as expected, with increasing m-value, the estimate's noise also increases. The results are summarized in table III as a function of the rotor speed and comb filter m-factor. The load angle is consistently underestimated during the transient period as visually apparent and reflected by the mean error μ . Remarkably, regardless of the used Fourier filter, the mean error μ remains more or less constant, except for $m=10$. Moreover, the standard deviation σ decreases consistently for all considered speeds upon switching from SDFT to GDFT with $m = 4$ or 6 . Beyond the tipping point where m equals 6, the standard deviation starts to rise again gradually. Finally, the SNR of the error signal consistently decreases with an increase in m-factor, indicating a deterioration in the estimation accuracy. Due to the reduced stability, higher estimation extremes are found. In figure 14, those extrema lines, indicating the maximum positive and negative load angle error that occurred, are added to the error signal plot.

2) *Evaluation of the percental speed gain:* An overview of the calculated time lags according to (15) is provided in table IV. For a speed of 1000 rpm, the determined lags and

TABLE III: The mean (μ), standard deviation (σ), signal-to-noise ratio (SNR), and the min/max instantaneous value of the error signal ϵ resulting from the difference between the estimated and measured load angles for a torque jump at different speeds depending on the used comb filter.

	0 - 0,2 Nm 1000 rpm					0 - 0,2 Nm 2000 rpm					0 - 0,2 Nm 3000 rpm				
	μ	σ	SNR	ϵ_{min}	ϵ_{max}	μ	σ	SNR	ϵ_{min}	ϵ_{max}	μ	σ	SNR	ϵ_{min}	ϵ_{max}
SDFT	-4,60	8,72	8,63	-31,01	19,04	-2,37	6,61	6,38	-22,70	19,24	-2,32	5,12	1,76	-21,42	16,95
GDFT _{m=4}	-4,41	6,28	-0,11	-26,37	19,98	-2,15	5,00	2,39	-22,05	17,41	-2,33	4,36	-1,84	-19,68	16,08
GDFT _{m=6}	-4,40	6,69	-5,29	-33,32	20,95	-2,10	4,89	0,17	-21,19	17,42	-2,32	4,31	-3,19	-18,35	16,55
GDFT _{m=8}	-4,39	8,40	-10,35	-44,45	31,36	-2,08	5,60	-2,56	-30,01	23,99	-2,32	4,88	-5,30	-21,98	18,15
GDFT _{m=10}	11,69	10,86	-9,79	-60,56	41,10	6,32	6,75	-5,00	-31,65	32,63	5,25	5,90	-7,75	-26,51	22,46

TABLE IV: Time delay between measured and estimated load angles, and resulting percentage speed gain compared to the classical SDFT algorithm, upon a torque jump.

	1000 tr/min 0 - 0,2 Nm		2000 tr/min 0 - 0,2 Nm		3000 tr/min 0 - 0,2 Nm		Expected time shift gain [%]
	Time shift [ms]	Time shift gain [%]	Time shift [ms]	Time shift gain [%]	Time shift [ms]	Time shift gain [%]	
SDFT	13,12	0	4,64	0	4,46	0	0
GDFT _{m=4}	6,96	46,95	2,74	40,95	3,22	27,80	50
GDFT _{m=6}	4,78	63,57	1,78	61,64	2,46	44,84	66,67
GDFT _{m=8}	3,46	73,63	1,56	66,38	2,64	40,81	75
GDFT _{m=10}	1,54	88,26	0,28	93,97	0,38	91,48	80

related percental gains of the different estimators are relatively close to the theoretically expected gains. However, as the speed increases, one notices that the percentage gains systematically decrease relative to expectations. Consequently, the estimator performs ‘worse’.

C. Experimental validation of the combined effect of an advanced Fourier filter with sampling strategy in the angular domain

The combined effect of different GDFT comb filter configurations and the renewed sampling strategy is analyzed by means of the previously described speed jump of 200 rpm in section V-B. As with the torque jump, the most instructive part is the transition from steady-state to transient and the ensuing load angle behavior. Therefore, figure 15 zooms in on the 20 ms before and 100ms after the velocity step from 1000 to 1200 rpm under no-load conditions.

1) *Evaluation of the accuracy and stability of the estimation:* At the bottom of figure 15, the instantaneous error ϵ is shown. Moreover, to get insight into the proposed estimators’ accuracy and stability, the mean μ , standard deviation σ , and SNR are summarized in table V for each of the considered filter designs. The mean error μ initially decreases with increasing initial speed (the speed before the speed jump), and rises again for an initial rotor speed of 3000 rpm compared to 2000 rpm. However, the average error does remain the largest at the lowest speed. On the other hand, the error’s spread σ does consistently decrease with an increment in base speed. The best average error μ for a particular speed jump profile is centered around $m=6$. Nevertheless, the mutual difference between the considered filters is small, and the results are invariably significantly better than for an SDFT filter. Finally, the SNR also lessens with a larger m -factor, indicating an increase in noise on the error signal and thus a more fluctuating estimate. The minimum ϵ_{min} and maximum ϵ_{max} estimation error, likewise summarized in table V, confirm previous indications. The extremes increase in absolute value, as expected, as

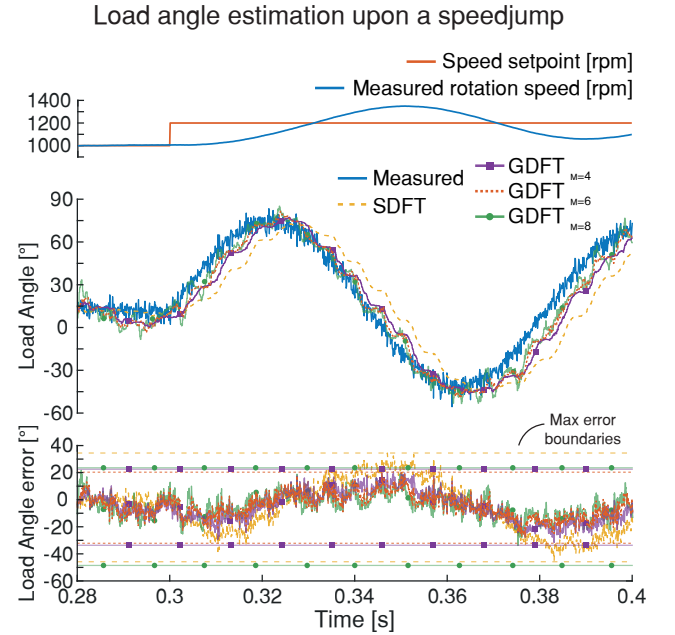


Fig. 15: Analysis of the estimated load angle $\hat{\delta}$ upon a speed step, based on the innovative GDFT structure with variable m -factor. The lower section shows the instantaneous error ϵ between the measured and estimated angles. Horizontal maximum extremity lines are added, indicating the maximum observed load angle error.

a comb filter with a higher m -value is used. On the contrary, the extremes decrease, like the spread, with an increase in initial speed.

2) *Evaluation of the percental speed gain:* As in section VI-B, the lag and percentage speed gain between the measured and estimated load angle are calculated based on equations (15) and (17). All results are summarized in table VI. Similar results were found at an initial speed of 1000 rpm for the

TABLE V: Average (μ), standard deviation (σ) and signal-to-noise ratio (SNR) of the error signal resulting from the instantaneous error between the calculated and measured load angle for a speed jump at different speeds depending on the used m-factor.

	1000 - 1200 rpm					2000 - 2200 rpm					3000 - 3200 rpm				
	0 Nm					0 Nm					0 Nm				
	μ	σ	SNR	ϵ_{min}	ϵ_{max}	μ	σ	SNR	ϵ_{min}	ϵ_{max}	μ	σ	SNR	ϵ_{min}	ϵ_{max}
SDFT	-6,45	3,83	12,68	-45,96	34,43	-5,13	3,03	9,53	-32,21	26,51	-5,46	2,68	6,76	-23,44	16,64
GDFT _{m=4}	-3,69	9,79	6,67	-33,79	22,47	-2,46	6,94	3,85	-26,97	20,06	-3,47	4,85	0,89	-20,21	13,44
GDFT _{m=6}	-3,71	8,03	1,99	-32,33	20,25	-2,47	6,07	0,12	-27,77	18,08	-3,45	4,45	-2,52	-19,12	13,59
GDFT _{m=8}	-3,75	8,18	-2,34	-48,71	23,53	-2,49	6,35	-3,28	-32,68	22,94	-3,46	4,91	-6,19	-23,88	18,41
GDFT _{m=10}	-3,80	9,35	-5,88	-77,30	30,93	-2,50	7,30	-6,30	-39,54	26,80	-3,46	5,87	-9,40	-28,48	19,08

TABLE VI: Time delay between measured and calculated load angles, and resulting percentage speedgain compared to the classical SDFT algorithm, upon a speed jump.

	1000 - 1200 tr/min		2000 - 2200 tr/min		3000 - 3200 tr/min		
	0 Nm		0 Nm		0 Nm		
	Time shift [ms]	Time shift gain [%]	Time shift [ms]	Time shift gain [%]	Time shift [ms]	Time shift gain [%]	Expected time shift gain [%]
SDFT	12,64	0	7,20	0	4,60	0	0
GDFT _{m=4}	6,40	49,37	3,36	53,33	1,68	63,48	50
GDFT _{m=6}	4,02	68,20	2,12	70,56	1,12	75,65	66,67
GDFT _{m=8}	2,94	76,74	1,40	80,56	0,36	92,17	75
GDFT _{m=10}	1,72	86,39	1,08	85,00	0,32	93,04	80

speed jump as for the torque jump. The difference between the observed and theoretically expected gain is minimal (1,53% for m=6). In contrast to the torque jump at higher speeds, the percentage speed gain becomes systematically bigger than the expected value. An incredible gain of 92% was observed for a comb filter with factor $m = 8$ at 3000 rpm, which is 17% above the theoretically expected percentage gain.

VII. CONCLUSION

This work explored whether the load angle, a measure of the motor operation efficiency, could be estimated with higher accuracy by using a generalized discrete Fourier filter in combination with a sampling strategy in the angular domain. Theoretically, it has been shown that a GDFT comb-filter with neutralizing effect on the 5th, 7th, 11th, 13th, ... harmonics makes an optimal trade-off between speed and accuracy and is, therefore, more efficient than the previously used SDFT filter. Measurement and analysis under transient operating conditions confirm the theoretical optimum. The average estimation error μ following a rotor velocity step almost halves in size by using a GDFT filter. For a torque step, it is not so much the average but rather the error's standard deviation σ that improves. The observed time gain on the estimate after a torque step for a filter with m equal to 6 is at least 45% and maximal 64% compared to an SDFT filter. For a speed jump, the observed gain varies between min 68% and max 75%. So, depending on the transient type, the result is better or worse than the theoretically expected profit of 66,67%. Across the different scenarios, the instantaneous error ϵ on the estimate at m=6 generally halves for a speed jump compared to the original SDFT filter, whereas for a torque jump, the standard deviation σ drops by roughly 25%.

Moreover, the measurements implicitly demonstrate that shifting from the time to the angular domain following the proposed method works and, as expected, has minimal impact on the load angle estimation. Variation in rotor speed combined with the spectral filter causes similar μ and σ trends throughout the transient for both speed and torque steps.

Therefore, in general, it can be concluded that the proposed load angle estimator succeeds in estimating the load angle more accurately, which directly opens up the possibility of more dynamic load angle control. Future research could evaluate the impact of estimated motor winding resistance R and inductance L parameters on the instantaneous load angle estimation error. Next, it is worth investigating the implementation of a GDFT post-processing filter to suppress the estimate's noise, and its effect on the instantaneous estimation error. Finally, this work's result gives rise to a re-examination of the dynamicity of the torque and velocity profiles that can be successfully controlled using the proposed estimator.

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conducting research on the sensorless control of BLDC and stepper motors.



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Jan Steckel graduated in electronic engineering at the University College "Karel de Grote" in Hoboken in 2007. In 2012 he obtained his doctoral degree at the University of Antwerp at the Active Perception Lab, with a dissertation titled "Array processing for in-air sonar systems - drawing inspirations from biology". During this period, he developed state-of-the-art sonar sensors, both biomimetic and sensor-array based. During his post-doc period he was an active member of the Centre for Care Technology at the University of Antwerp where he was in charge of various healthcare-related projects concerning novel sensor technologies. Furthermore, he pursued industrial exploitation of the patented 3D array sonar sensor which was developed in collaboration during his Ph.D. In 2015 he became a tenure track professor at the University of Antwerp in the Constrained Systems Lab where he researches sensors, sensor arrays, and signal processing algorithms using an embedded, constrained systems approach.



optimization of mechatronic systems.

Stijn Derammelaere was born in Kortrijk, Belgium, in 1984. He received the Master degree in automation from the Technical University College of West-Flanders, Belgium, in 2006. He received the Ph.D. degree in 2013 at Ghent University where he initiated research concerning control engineering, co-design of control and mechatronic systems and the motion control of fractional horsepower machines. He continues this research since 2017 at the University of Antwerp as a professor in mechatronics. His expertise includes co-design, motion control and